

LAB # 06

Open Ended Lab

Objective

Implementation of open ended lab.

Lab Goals

To provide working insight into the usage of operators, conditional statements, loop, math class, random class in java programming:

- a. Conditional Statement.
- b. Operators
- c. Loop
- d. Math Class
- e. Random Class

Exercise

- A. **Given Scenario:** You are developing a simple Java application for a mathematics quiz game. In this quiz, the player will be asked to solve basic arithmetic problems such as addition, subtraction, multiplication, and division. You want to implement a loop to keep the game running until the player decides to exit. Additionally, you want to include a condition where the player can choose the difficulty level of the questions. To achieve this, you decide to use the Math class to generate random numbers for the questions.

Task: To solve the scenario follow the below given guidelines:

1. Write a Java program that implements the mathematics quiz game with the following specifications:
2. The program should display a menu to the player where they can choose the difficulty level of the questions (Easy, Medium, or Hard).
3. Based on the difficulty level chosen by the player, the program should generate random arithmetic questions accordingly.
4. After each question, the player should be prompted to enter their answer.
5. If the player's answer is correct, display a message congratulating them; otherwise, display a message indicating the correct answer.
6. After each question, ask the player if they want to continue playing. If they choose to continue, present them with another question; if not, exit the game.
7. Use loops to keep the game running until the player decides to exit.

Your task is to write the Java code to implement the above specifications.

Your output is just like:

```
Welcome to Math Quiz Game!
Choose Difficulty Level:
1. Easy
2. Medium
3. Hard
Enter your choice: 2
45 * 27 = ?
1215
Congratulations! Your answer is correct.
Do you want to play again? (yes/no): yes
Choose Difficulty Level:
1. Easy
2. Medium
3. Hard
Enter your choice: 4
Invalid choice. Setting difficulty to Easy.
6 / 2 = ?
3
Congratulations! Your answer is correct.
Do you want to play again? (yes/no): no
Thank you for playing Math Quiz Game!
```